VECTOR

Workflow Overview

Technical Reference

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Reference Documents

No.	Source	Title	Version
[1]			



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1 Introduction

The **DaVinci Configurator** provides a comprehensive workflow for the update of a DaVinci project configuration including DaVinci Developer.

What is the **intention** of such update workflow?

> User may update the DaVinci project in case of modifications in the input files without deleting manual configurations

> Ensures working on the newest data model

When is it necessary to update the project?

> Setting up a project with input files from OEM

> Getting new input files from OEM

> Special use-case: Obtaining a new SIP

The update workflow is split into two parts: **File Preprocessing** and **Project Update**

Project Update Workflow can be started either from the DaVinci Configurator GUI (Input File Assistant), or from the command line tool. The Workflow Overview can be seen below on Figure 1-1.

Note
The selection of input files can be done with the Input File Assistant of DaVinci Configurator. Please refer to online help for details.

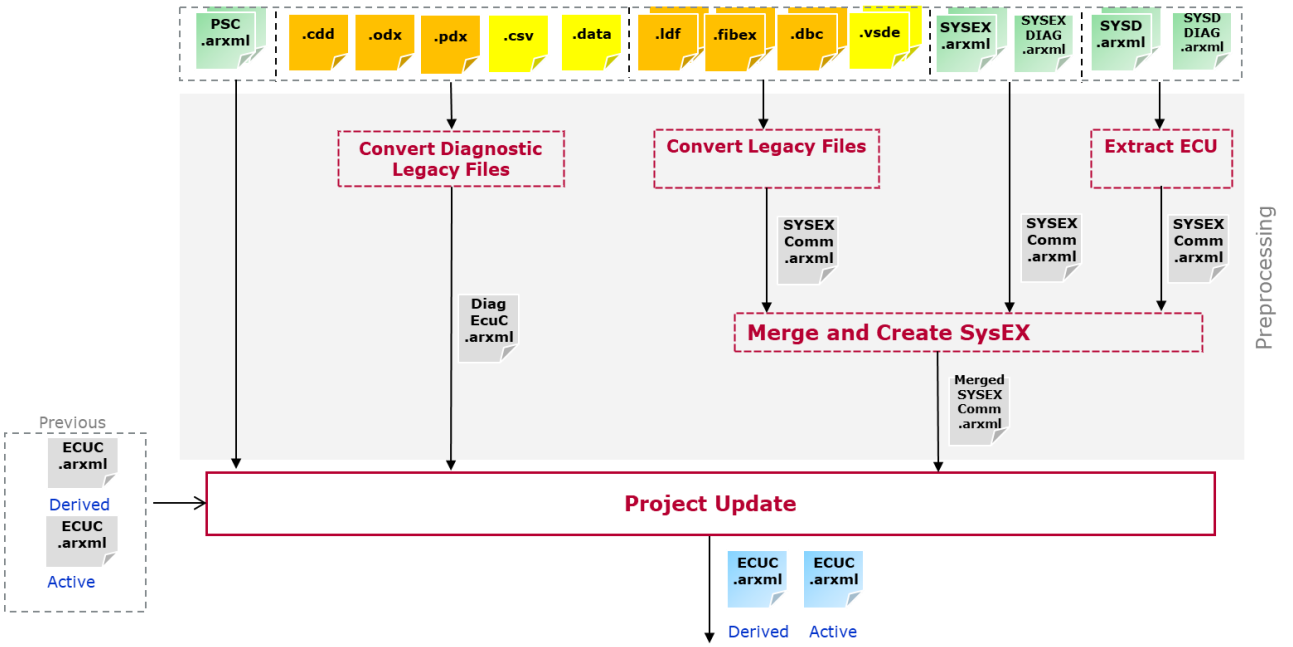


Figure 1-1 Workflow Overview

2 File Preprocessing Phase

On Figure 2-1 can be seen an overview of only the Preprocessing phase. Below are the steps involved in it.

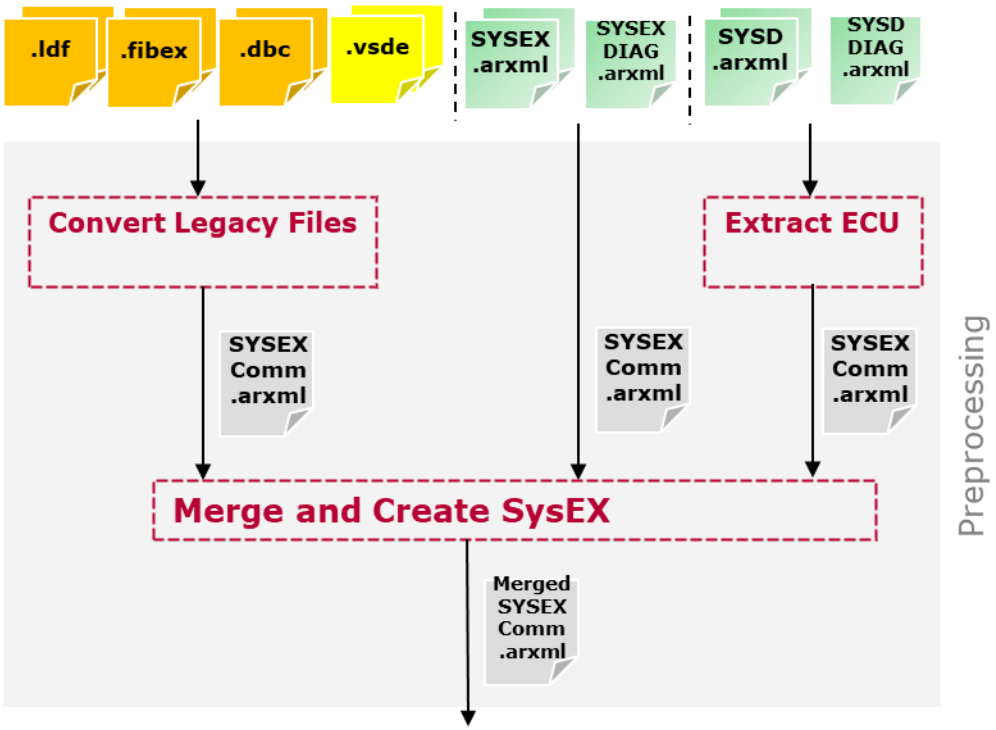


Figure 2-1 Preprocessing

- **Communication Legacy file** has to be converted into ARXML files by Vector Legacy Converter
- **System Extract** doesn't have to be converted
- **System Description** has to be extracted into a System Extract of one ECU
- Converted **Communication Legacy files**, imported **System Extract** and extracted **System Description** will be merged into one **System Extract ARXML**
- These files (Communication .arxml and SystemExtract) can be found at the Project Folder "<Project>\Config\System".
- **Merge and Create System Extract** consists of several steps to merge **multiple input files into one System Extract**

  a. The following are searched through the InputFiles for their Merge:
     i. EcuInstance
     ii. System
     iii. RootSwCompositionPrototype
     iv. CompositionType
  b. Package structure is taken from an input file to integrate all other objects into this structure in a generic way

The merge performs an additive merge/overlay of the AUTOSAR models based on the AUTOSAR package structure. As long as there is no conflicting information, the User can add as many of these files as he wants, and the DaVinci Configurator would be able to merge it.



Figure 2-2 Mergeable Files

The Following **example** shows the limits of the functionality:

- A converted Communication Legacy File and an OEM Extract shall be merged. The LegacyFileConverter always creates a fix package structure. This structure most likely differs from the structure of the OEM Extract.
- The functionality cannot merge all information for the same object until the package structure of the converted legacy file and OEM Extract is identical

2.1 Project Standard Configuration (PSC) addition Example:

On a PSC file, a COMM Module may be described.

- This module can be extended by another PSC file, as long as keeps the Module Structure identical (See Figure 6 –ComM Main Nodes).
  - In this case, on “Containers” are new, but the rest (UUID, Implementation-config-variant, module-description ref) shall keep the same value, or an error will show up.
- As an overall rule, files must be splitted according to the Split Rules & the attributes need to be consistent in all files.

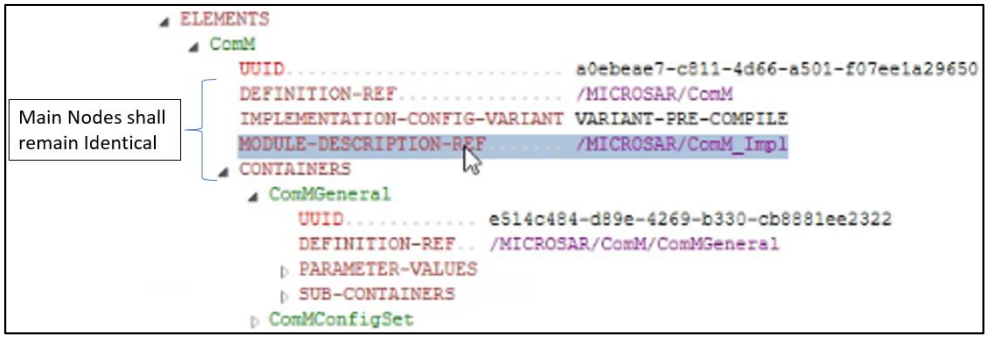


Figure 2-3 Module Structure

2.2 A Second Common Example (Adding CAN):

Adding a new CAN channel doesn’t require the same structure, since in Autosar terms a new CAN channel is a new Can-Cluster. Therefore, a new CAN cluster with a new Physical Channel would have to be added.

```
AR-PACKAGE Cluster
  ELEMENTS
    CAN-CLUSTER CAN
      LONG-NAME
      ADMIN-DATA
      CAN-CLUSTER-VARIANTS
        CAN-CLUSTER-CONDITIONAL
          BAUDRATE ..... 500000
          PHYSICAL-CHANNELS
            CAN-PHYSICAL-CHANNEL CHNL
              COMM-CONNECTORS
              FRAME-TRIGGERINGS
              I-SIGNAL-TRIGGERINGS
              PDU-TRIGGERINGS
          PROTOCOL-NAME .... CAN
      REF-BY SYSTEM ..... /VehicleProject/System
      REF-BY CAN-NM-CLUSTER /NM/NmConfig/CAN
      REF-BY CAN-TP-CONFIG  /TP/CAN
```

Figure 2-4 Can Cluster

2.3 Non-Mergeable Files

In contrast to ARXMLs, Legacy Diagnostics cannot be merged together (only one file per DPA is allowed), this is due to the capability of the Converter tool that is being used. The result of this tool is an ECUC Description.

While the conversion is done on the Preprocessing phase, the results is used on the Project Update Phase.



Figure 2-5 Non-Mergeable Files

3 Project Update

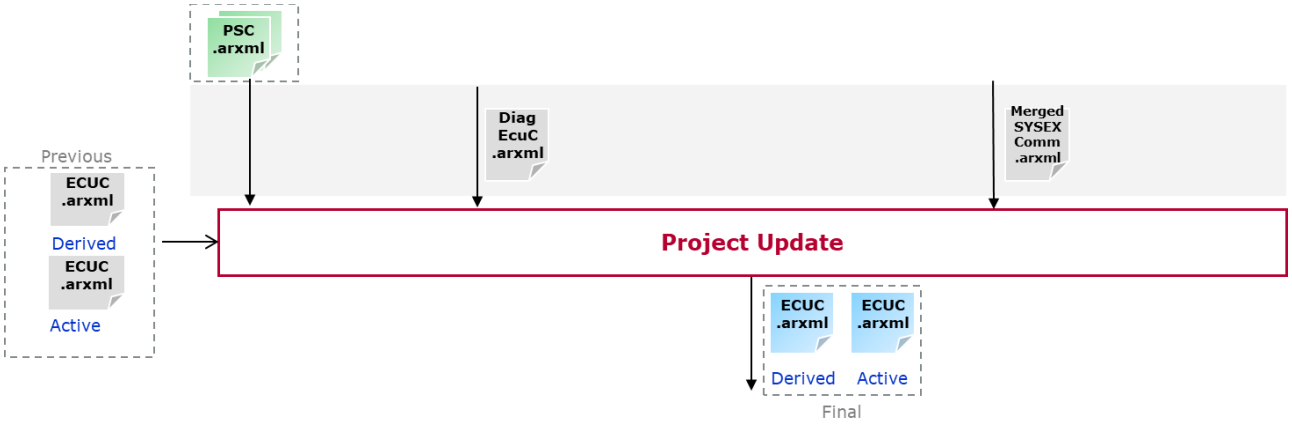


Figure 3-1 Project Update

The **Project Update** consists of several steps that are executed automatically step-by-step.

- **BaseEcuC Generator** creates initial configuration (InitialEcuC) based on (new) input files. This is the outcome of all derivation rules. After that, the the preconfiguration and recommended configuration is applied. The result in the BaseEcuC.
- **EcuC Update** updates the active configuration (ActiveEcuC) with the Base configuration (The ActiveEcuC is initially a copy of the InitialEcuc). Changes, once saved, are represented on the ActiveEcuc. In other words, the Base configuration is merged into the current project configuration.
- **Execute Solving Actions** Executes solving actions of the MICROSAR module validation that normally would need to be executed by the user. As a result, the Active EcuC will be completed and the Service SWCs and BSW Internal Behavior will be updated.
- **Developer Update** updates DaVinci Developer workspace (based on created System Extract)
- **RTE Update** synchronizes the new flattened Ecu Extract and executes automatic RTE setup

3.1 Additional Details

Pre-Configuration and Recommended values come from the SIP and are applied automatically.

3.1.1 Pre-configuration

When the DaVinci Configurator loads a project, it initially applies all pre-configuration values (instance and class pre-configuration) to the loaded model before allowing the user configuring the project. All pre-configuration values are read-only then. The user is not allowed to change them. The Pre-configuration values are applied at the right moment to avoid conflicts.

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3.1.2 Conflicts Between Pre-Configuration and Derived Values

When the BaseEcuC generator finally overwrites the configuration values with the pre-configuration, the pre-configuration shall have the higher priority. If a derived value differs from the pre-configuration, it shall be overwritten. In this case the user shall be informed to allow her patching the system extract accordingly.

3.1.3 Recommended and Default Values

The DaVinci Configurator applies recommended and default values not automatically after loading a project but always when a new object is being created. When e.g. a container is being created which has a related recommended configuration, all contained sub-containers and parameters of the recommended configuration are created too. The behavior of the BaseEcuC generator shall be the same here.

Lastly, in comparison to the Preconfiguration, preconfiguration has a higher priority, therefore those values would remain.

3.1.4 Base Ecuc Workflow Execution Sequence

Based on the Base Ecuc workflow described in Figure 7, the following control flow will be executed:

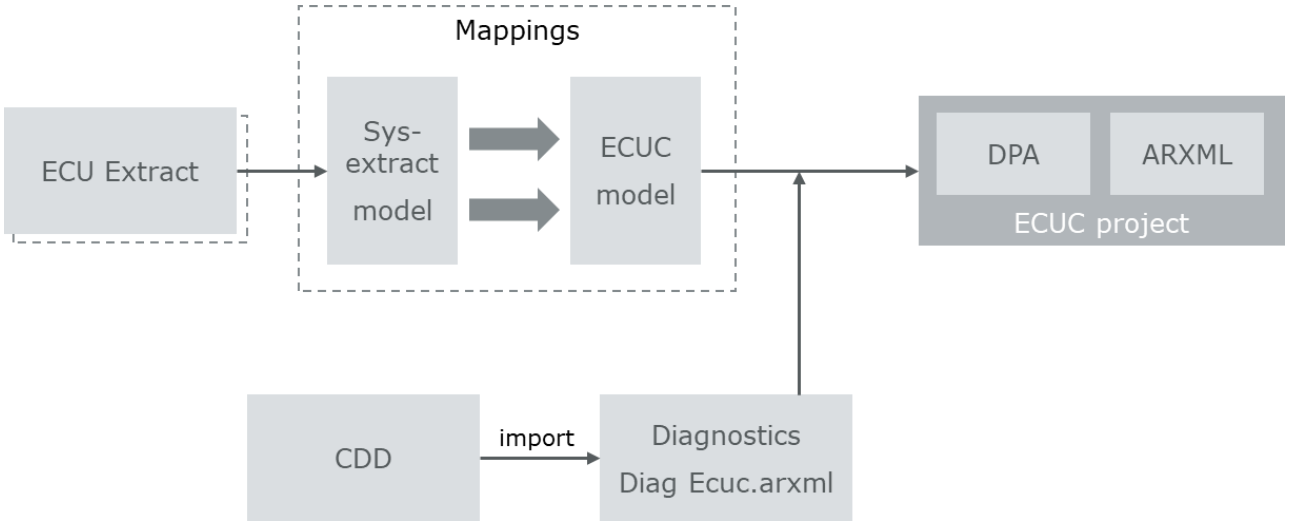


Figure 3-2 Base EcuC Workflow

1. Load DPA project
   - > Provides an empty ECUC model
   - > Contains all relevant BSW module definitions contained in the SIP
   - > DPA contains the file names of the required extract files
2. Load the required extract file(s)
3. Execute mappings and create the internal ECUC model
4. Apply recommended and default values

5. Import CDD (Diagnostics)
   > Convert to ECUC File
6. Apply pre-configuration values
7. Create missing UUIDs in the ECUC model

3.2 EcuC Update Workflow

An updated project is the result of a model merge based on the current project and a new created one. The purpose is to keep the user's modifications made in the current project and merge them into the new one where possible.

An example where this is required is when the OEM provides an updated system extract.

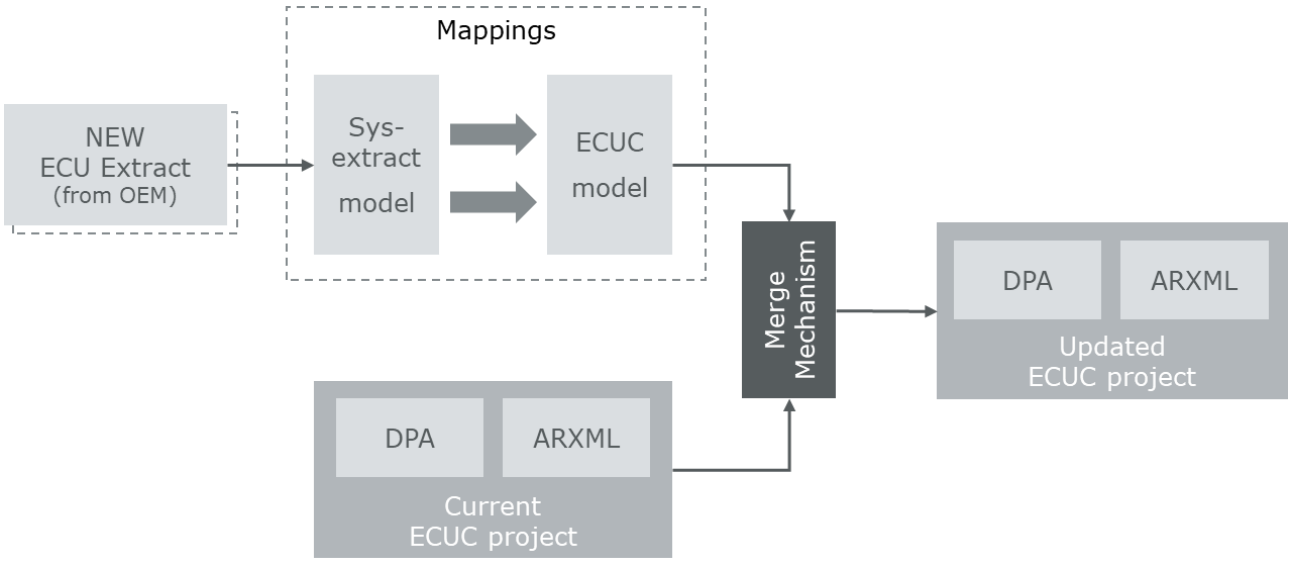


Figure 3-3 Updated EcuC Project

The **DaVinci Configurator** supports different update options with different number of sub-steps to be executed depending on the use-case.

> Update of ECU configuration and DaVinci Developer workspace

  > If the **merged System Extract** has been modified in a way of concerning also DaVinci Developer workspace.

Note
This is a safe default option if the user is not sure what kind of changes have been made to the input files.

> Update of ECU configuration only

- If **no DaVinci Developer workspace** is available or the modifications in the merged System Extract do not concern the DaVinci Developer workspace.

> Update Legacy Diagnostic Data files only

- If only **Legacy Diagnostic Data** files has been modified (.cdd, .odx). (Note: Diagnostic Extract not considered)

4 Questions and Answers

4.1 What is the effect if I completely remove an input file?

All what was previously derived from that Input File would be deleted. This includes also potential changes to the derived elements made by the user (e.g. further parameters that have been added).

4.2 What happens if we change the name of an object (for example, a PDU)?

There are 2 cases. There could be a simple model without UUIDs, or one with the use of UUIDs.



Note
With an according setting in the DPA project file you can force the DaVinci Configurator to ignore the UUIDs in the input files. See online help for details..

> If no UUIDs are used: Once the name is changed on the Input File, the object/PDU will be considered a new PDU. Since the Old name PDU is no longer found, it will be deleted including any derivations that came out of it. Since it now has a new name it will be considered a new object, and the normal derivations out of it will be newly created. In other words, any changes made on parameters derived from the original-named PDU will be lost.

> If UUIDs are used: Only the name will change, as the UUID is used to identify all relationships of the PDU.

4.3 What if we add a new PDU?

At least a global PDU is created for it, the rest of the derivations beyond depends on the PDU type. The new PDU will have be no conflict with other PDUs.



Reference
For more details on PDUs and their Types, see AUTOSAR_TPS_SYSTEMTEMPLATE, Chapter 6 Communication.

4.4 What happens if the cycle time of a CAN message is changed?

The ECUC will update the parameter unless it is a User Defined parameter.

4.5 What are User-Defined Parameters?

Parameter which are annotated as **User-Defined** are not being changed during the Database update. They will be overtaken into the Active ECUC file as they are. But the Initial ECUC file will be contain the value derived from the input files.

4.6 What options are there to Import ECUC files?

- **A Project Standard Configuration (**PSC): this file allows the OEM to deliver project-specific pre-configurations of ECUC elements. It is used for diagnostic data and OEM-specific components.
  - Imported as derived elements (read-only)
  - PSC files overwrite whatever is already written in the module.
- **Module Import:** Import of an ECUC file containing a module configuration. Imported as read-write.
- **ECUC Files References**: ECUC files which are shared with other projects. Imported into the project as read-only. Import is with the Input Files Assistant is not possible.
  - If the files are changed in another project, then the changes will be synchronized
  - If you have a 3rd party project where there is already some configurated data, this can be included and would remain READ ONLY. The difference between PSC and External ECU files is that it can only be used if there is no module already present. This method doesn't overwrite, it only makes its configuration visible (as Read Only).

4.7 Is Project Update also needed during a SIP update, even though the Input files did not change?

Yes, in newer SIPs there are is usually a new DaVinci Configurator Version, or some other external tools. These new versions may contain new or modified derivations. The external tools also may influence on the derivations. Lastly, the definition of a BSW module can also be changes, which can only be remedied by a Project Update.

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5 Glossary and Abbreviations

5.1 Glossary

Term	Description

5.2 Abbreviations

Abbreviation	Description
BSW	Basic Software
PSC	Project Standard Configuration
PDU	Protocol Data Unit

6 Contact

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